

```
from google.colab import files
```

```
uploaded = files.upload()
```



Choose files

 netflix.csv

- **netflix.csv**(text/csv) - 3399671 bytes, last modified: 17/06/2024 - 100% done

Saving netflix.csv to netflix.csv

```
import numpy as np # linear algebra operations
import matplotlib.pyplot as plt
import seaborn as sns
import io
import pandas as pd # used for data preparation
import plotly.express as px #used for data visualization
from textblob import TextBlob #used for sentiment analysis
```

```
df = pd.read_csv(io.BytesIO(uploaded['netflix.csv']))
print(df)
```



|      |       |         |             |                 |
|------|-------|---------|-------------|-----------------|
| 8803 | s8804 | TV Show | Zombie Dumb | NaN             |
| 8804 | s8805 | Movie   | Zombieland  | Ruben Fleischer |
| 8805 | s8806 | Movie   | Zoom        | Peter Hewitt    |
| 8806 | s8807 | Movie   | Zubaan      | Mozez Singh     |

|      |                                                   | cast | country \     |
|------|---------------------------------------------------|------|---------------|
| 0    |                                                   | NaN  | United States |
| 1    | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... |      | South Africa  |
| 2    | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... |      | NaN           |
| 3    |                                                   | NaN  | NaN           |
| 4    | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... |      | India         |
| ...  |                                                   | ...  | ...           |
| 8802 | Mark Ruffalo, Jake Gyllenhaal, Robert Downey J... |      | United States |
| 8803 |                                                   | NaN  | NaN           |
| 8804 | Jesse Eisenberg, Woody Harrelson, Emma Stone, ... |      | United States |
| 8805 | Tim Allen, Courteney Cox, Chevy Chase, Kate Ma... |      | United States |
| 8806 | Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan... |      | India         |

|   | date_added         | release_year | rating | duration \ |
|---|--------------------|--------------|--------|------------|
| 0 | September 25, 2021 | 2020         | PG-13  | 90 min     |
| 1 | September 24, 2021 | 2021         | TV-MA  | 2 Seasons  |
| 2 | September 24, 2021 | 2021         | TV-MA  | 1 Season   |

```

                                listed_in \
0                                Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2    Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV
4    International TV Shows, Romantic TV Shows, TV ...
...                                ...
8802                                Cult Movies, Dramas, Thrillers
8803                                Kids' TV, Korean TV Shows, TV Comedies
8804                                Comedies, Horror Movies
8805                                Children & Family Movies, Comedies
8806    Dramas, International Movies, Music & Musicals

                                description
0    As her father nears the end of his life, filmm...
1    After crossing paths at a party, a Cape Town t...
2    To protect his family from a powerful drug lor...
3    Feuds, flirtations and toilet talk go down amo...
4    In a city of coaching centers known to train I...
...                                ...
8802    A political cartoonist, a crime reporter and a...
8803    While living alone in a spooky town, a young g...
8804    Looking to survive in a world taken over by zo...
8805    Dragged from civilian life, a former superhero...
8806    A scrappy but poor boy worms his way into a ty...

[8807 rows x 12 columns]

```

## ✓ Checking number of rows and columns in data

`df.shape`

 (8807, 12)

## ✓ Checking content available in Dataset

`df.head()`



|   | show_id | type    | title                 | director        | cast                                              | country       | date_added         | release_ye |
|---|---------|---------|-----------------------|-----------------|---------------------------------------------------|---------------|--------------------|------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson | NaN                                               | United States | September 25, 2021 | 20         |
| 1 | s2      | TV Show | Blood & Water         | NaN             | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 20         |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 | 20         |
| 3 | s4      | TV Show | Jailbirds New Orleans | NaN             | NaN                                               | NaN           | September 24, 2021 | 20         |
| 4 | s5      | TV Show | Kota Factory          | NaN             | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 | 20         |

Next steps:

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```
df.head()
```



|   | show_id | type    | title                 | director        | cast                                              | country       | date_added         | release_ye |
|---|---------|---------|-----------------------|-----------------|---------------------------------------------------|---------------|--------------------|------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson | NaN                                               | United States | September 25, 2021 | 20         |
| 1 | s2      | TV Show | Blood & Water         | NaN             | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 20         |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 | 20         |
| 3 | s4      | TV Show | Jailbirds New Orleans | NaN             | NaN                                               | NaN           | September 24, 2021 | 20         |
| 4 | s5      | TV Show | Kota Factory          | NaN             | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 | 20         |

✓ **How to check columns name of dataset**

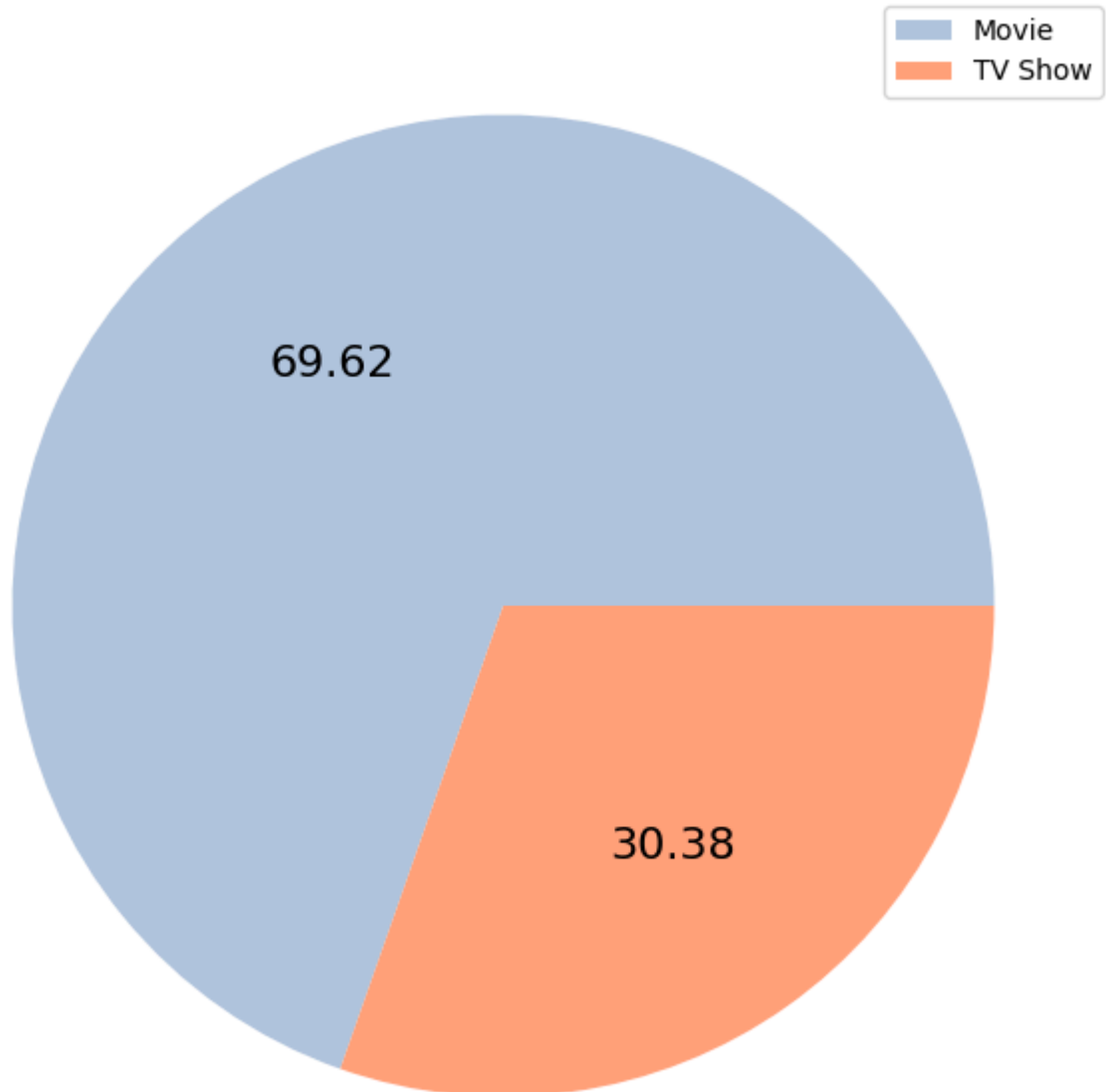
```
df.columns
```

Index(['show\_id', 'type', 'title', 'director', 'cast', 'country', 'date\_added', 'release\_year', 'rating', 'duration', 'listed\_in', 'description'], dtype='object')

✓ **Checking the type of the content**

```
plt.figure(figsize=(10,8))

plt.pie(df.type.value_counts(),
        labels = df.type.value_counts().index,
        labeldistance = None, autopct="%.2f",
        textprops = {'fontsize': 16,},
        colors = ['lightsteelblue','lightsalmon' ] )
plt.legend()
plt.show()
```



Result: The number of movie content on netflix is 2.33 times the TV Shows.

## ✓ Taking the count of ratings available

```
x = df.groupby(['rating']).size().reset_index(name='counts')
print(x)
```



|    | rating   | counts |
|----|----------|--------|
| 0  | 66 min   | 1      |
| 1  | 74 min   | 1      |
| 2  | 84 min   | 1      |
| 3  | G        | 41     |
| 4  | NC-17    | 3      |
| 5  | NR       | 80     |
| 6  | PG       | 287    |
| 7  | PG-13    | 490    |
| 8  | R        | 799    |
| 9  | TV-14    | 2160   |
| 10 | TV-G     | 220    |
| 11 | TV-MA    | 3207   |
| 12 | TV-PG    | 863    |
| 13 | TV-Y     | 307    |
| 14 | TV-Y7    | 334    |
| 15 | TV-Y7-FV | 6      |
| 16 | UR       | 3      |

## ✓ Creating the Piechart based on Content rating

```
new_catgs = {
    'TV-PG': 'Parental Guidance',
    'TV-MA' : 'Mature Audience',
    'TV-Y7-FV': 'Teens',
    'TV-Y7': 'Teens',
    'TV-14': 'Teens',
    'R': 'Mature Audience',
    'TV-Y': 'General Audience',
    'NR': 'Mature Audience',
    'PG-13': 'Teens',
    'TV-G': 'General Audience',
    'PG': 'Teens',
    'G': 'General Audience',
    'UR': 'Mature Audience',
    'NC-17': 'Mature Audience'
}
df['rating']=df['rating'].replace(new_catgs)
df.head()
```



|   | show_id | type    | title                 | director        | cast                                              | country       | date_added         | release_ye |
|---|---------|---------|-----------------------|-----------------|---------------------------------------------------|---------------|--------------------|------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson | NaN                                               | United States | September 25, 2021 | 20         |
| 1 | s2      | TV Show | Blood & Water         | NaN             | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 20         |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 | 20         |
| 3 | s4      | TV Show | Jailbirds New Orleans | NaN             | NaN                                               | NaN           | September 24, 2021 | 20         |
| 4 | s5      | TV Show | Kota Factory          | NaN             | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 | 20         |

Next steps:

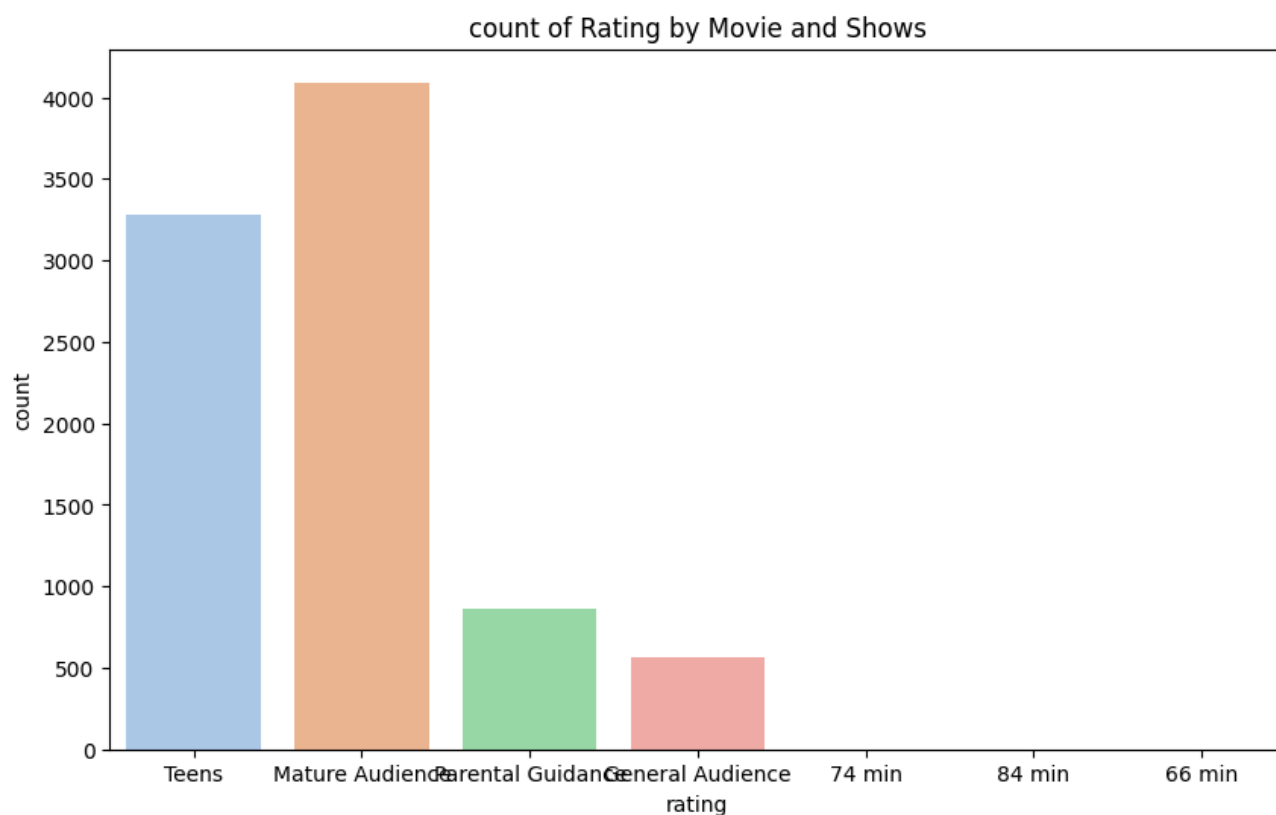
[Generate code with df](#)

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```
plt.figure(figsize= (10,6))
sns.countplot(x="rating", data=df, palette="pastel",)
plt.title("count of Rating by Movie and Shows");
```

 <ipython-input-37-10d770d36b2d>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in



Result: It is seen that Netflix has more content for Mature audience.

## ✓ Analyzing the top 5 Directors on Netflix

```
df['director']=df['director'].fillna('Director not specified')  
df.head()
```





|   | show_id | type    | title                 | director               | cast                                              | country       | date_added         | release_ye |
|---|---------|---------|-----------------------|------------------------|---------------------------------------------------|---------------|--------------------|------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson        | NaN                                               | United States | September 25, 2021 | 20         |
| 1 | s2      | TV Show | Blood & Water         | Director not specified | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 20         |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq        | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 | 20         |
| 3 | s4      | TV Show | Jailbirds New Orleans | Director not specified | NaN                                               | NaN           | September 24, 2021 | 20         |
| 4 | s5      | TV Show | Kota Factory          | Director not specified | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 | 20         |

Next steps:

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```
directors_list = pd.DataFrame()
print(directors_list)
```



Empty DataFrame  
Columns: []  
Index: []

```
directors_list = df['director'].str.split(',', expand=True).stack()
print(directors_list)
```



```
0      0      Kirsten Johnson
1      0  Director not specified
2      0      Julien Leclercq
3      0  Director not specified
4      0  Director not specified
...
8802  0      David Fincher
8803  0  Director not specified
```

```

8804 0          Ruben Fleischer
8805 0          Peter Hewitt
8806 0          Moez Singh
Length: 9612, dtype: object

```

```

directors_list = directors_list.to_frame()
print(directors_list)

```

```

0          0          Kirsten Johnson
1          0  Director not specified
2          0          Julien Leclercq
3          0  Director not specified
4          0  Director not specified
...
8802 0          David Fincher
8803 0  Director not specified
8804 0          Ruben Fleischer
8805 0          Peter Hewitt
8806 0          Moez Singh

```

[9612 rows x 1 columns]

```

directors_list.columns = ['Director']
print(directors_list)

```

```

0          0          Director
1          0  Director not specified
2          0          Julien Leclercq
3          0  Director not specified
4          0  Director not specified
...
8802 0          David Fincher
8803 0  Director not specified
8804 0          Ruben Fleischer
8805 0          Peter Hewitt
8806 0          Moez Singh

```

[9612 rows x 1 columns]

```

directors = directors_list.groupby(['Director']).size().reset_index(name='Total C
print(directors)

```

```

0          Director  Total Count
1          Aaron Moorhead          2
2          Aaron Woolf          1
3  Abbas Alibhai Burmawalla          1
4          Abdullah Al Noor          1
5          Abhinav Shiv Tiwari          1
...
5116          Çagan Irmak          1
5117          Ísold Uggadóttir          1
5118          Óskar Thór Axelsson          1
5119          Ömer Faruk Sorak          2
5120          Şenol Sönmez          2

```

[5121 rows x 2 columns]

Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.

```
directors = directors[directors.Director != 'Director not specified']
print(directors)
```

```
↵
      Director  Total Count
0      Aaron Moorhead      2
1      Aaron Woolf      1
2  Abbas Alibhai Burmawalla      1
3      Abdullah Al Noor      1
4      Abhinav Shiv Tiwari      1
...
5116      Çagan Irmak      1
5117      Ísold Uggadóttir      1
5118      Óskar Thór Axelsson      1
5119      Ömer Faruk Sorak      2
5120      Şenol Sönmez      2
```

[5120 rows x 2 columns]

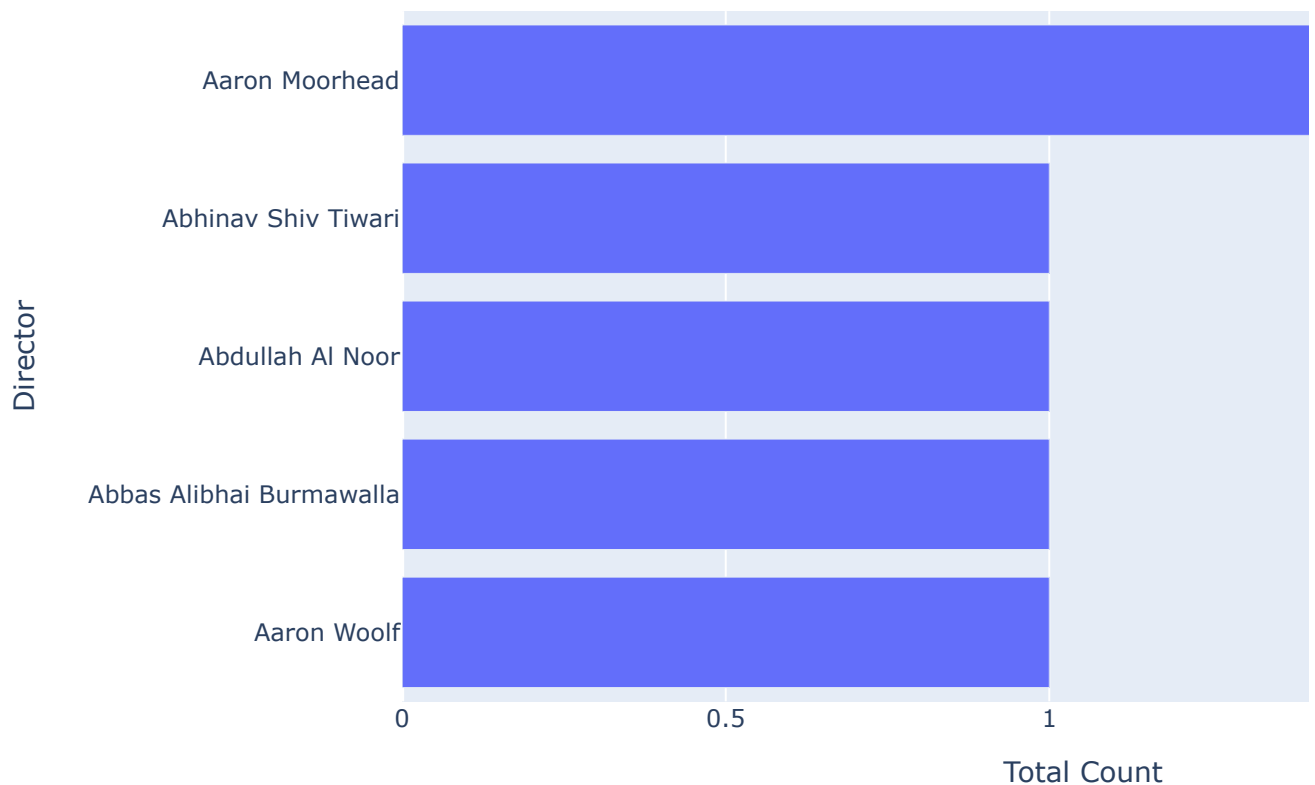
```
top5Directors = directors.head()
print(top5Directors)
```

```
↵
      Director  Total Count
0      Aaron Moorhead      2
1      Aaron Woolf      1
2  Abbas Alibhai Burmawalla      1
3      Abdullah Al Noor      1
4      Abhinav Shiv Tiwari      1
```

```
top5Directors = top5Directors.sort_values(by=['Total Count'])
barChart = px.bar(top5Directors, x='Total Count', y = 'Director', title = 'Top 5
barChart.show()
```



## Top 5 Directors on Netflix



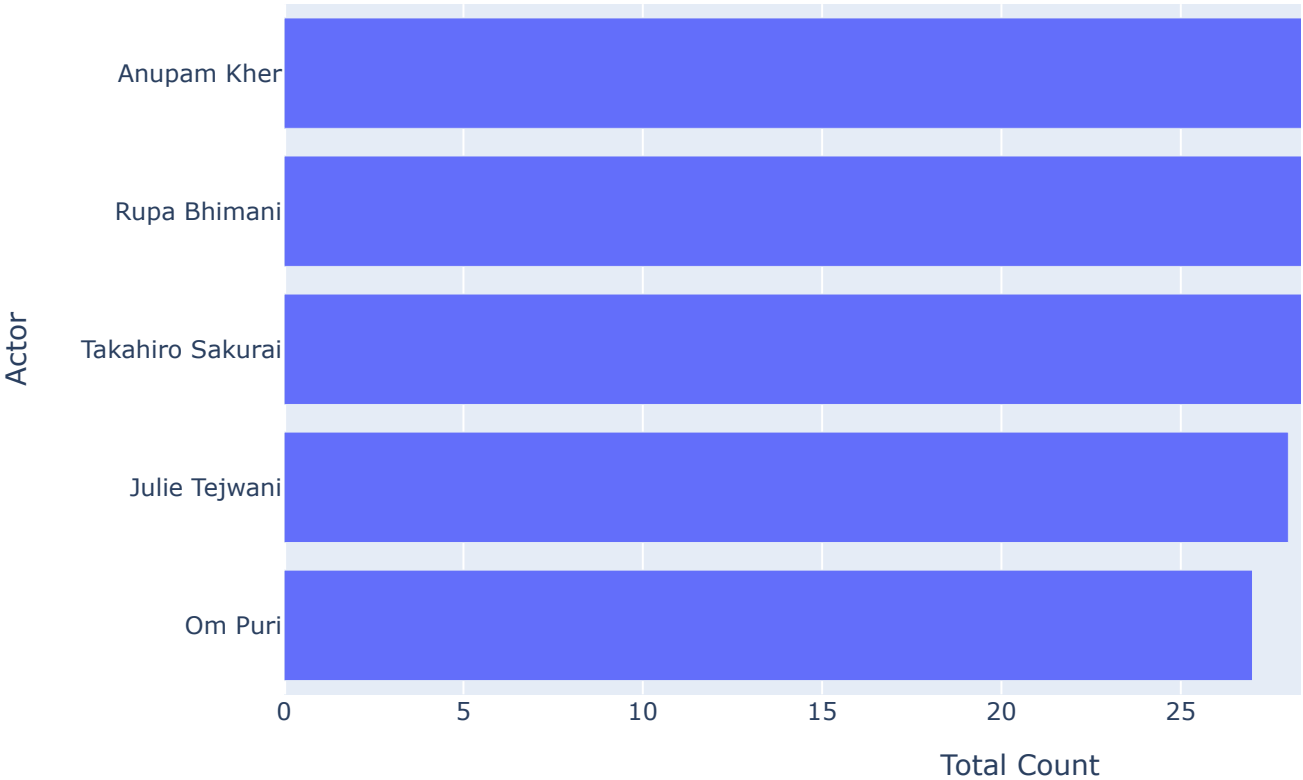
Result: These are the top 5 directors who has directed the most in Netflix

## ✓ Analyzing the top 5 Actors in Netflix

```
df['cast']=df['cast'].fillna('No cast specified')
cast_df = pd.DataFrame()
cast_df = df['cast'].str.split(',',expand=True).stack()
cast_df = cast_df.to_frame()
cast_df.columns = ['Actor']
actors = cast_df.groupby(['Actor']).size().reset_index(name = 'Total Count')
actors = actors[actors.Actor != 'No cast specified']
actors = actors.sort_values(by=['Total Count'], ascending=False)
top5Actors = actors.head()
top5Actors = top5Actors.sort_values(by=['Total Count'])
barChart2 = px.bar(top5Actors, x='Total Count', y='Actor', title='Top 5 Actors on
barChart2.show()
```



### Top 5 Actors on Netflix



Result: These are the top 5 actors in Netflix

```
df.head()
```



|   | show_id | type    | title                 | director               | cast                                              | country       | date_added         | release_ye |
|---|---------|---------|-----------------------|------------------------|---------------------------------------------------|---------------|--------------------|------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead  | Kirsten Johnson        | No cast specified                                 | United States | September 25, 2021 | 20         |
| 1 | s2      | TV Show | Blood & Water         | Director not specified | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 20         |
| 2 | s3      | TV Show | Ganglands             | Julien Leclercq        | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN           | September 24, 2021 | 20         |
| 3 | s4      | TV Show | Jailbirds New Orleans | Director not specified | No cast specified                                 | NaN           | September 24, 2021 | 20         |
| 4 | s5      | TV Show | Kota Factory          | Director not specified | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India         | September 24, 2021 | 20         |

Next steps:

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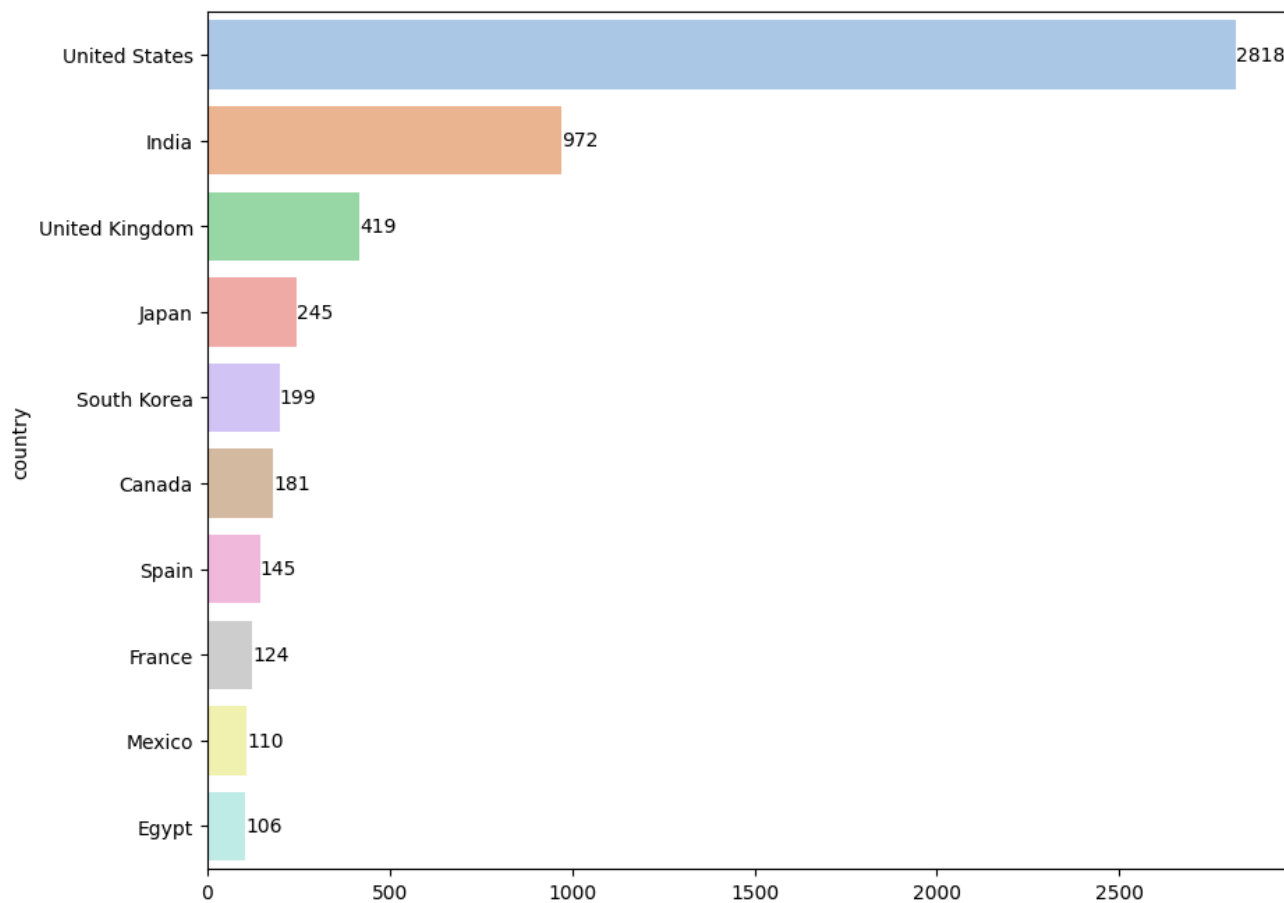
## ✓ Analyzing the top 10 countries in Netflix

Double-click (or enter) to edit

```
top_10_countries= df.country.value_counts().head(10)
top_10_countries = pd.DataFrame(top_10_countries)
top_10_countries
plt.figure(figsize = (10,8))
country_plot = sns.barplot(x = df.country.value_counts()[ :10].values,
                           y= df.country.value_counts()[ :10].index,palette = "pas
for i in country_plot.containers:
    country_plot.bar_label(i);
```

 <ipython-input-32-d206729b2be3>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in



Result: About 32% of content on Netflix is produced in the USA.

✓ **Analyzing the trend of content produced on Netflix every year**

```
df1 = df[['type', 'release_year']]
df1 = df1.rename(columns = {"release_year": "Release Year", "type": "Type"})
df2 = df1.groupby(['Release Year', 'Type']).size().reset_index(name='Total Count')
print(df2)
```

```

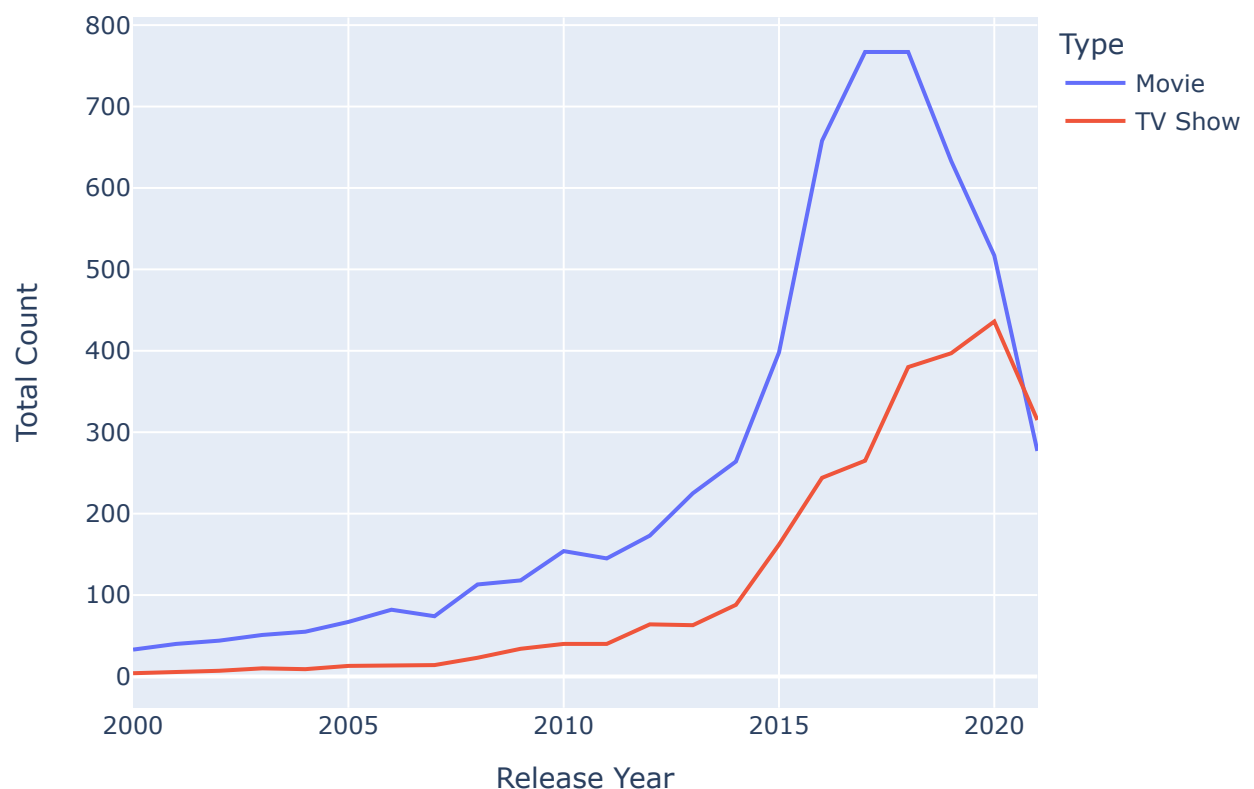
➡
   Release Year  Type  Total Count
0         1925  TV Show           1
1         1942   Movie           2
2         1943   Movie           3
3         1944   Movie           3
4         1945   Movie           3
..          ...    ...          ...
114        2019  TV Show        397
115        2020   Movie        517
116        2020  TV Show        436
117        2021   Movie        277
118        2021  TV Show        315
```

[119 rows x 3 columns]

```
df2 = df2[df2['Release Year']>=2000]
graph = px.line(df2, x = "Release Year", y="Total Count", color = "Type", title =
graph.show())
```



### Trend of Content Produced on Netflix Every Year





Result: It is seen that most content on netflix was released in year 2018, the amount of content is

## ✓ Sentiment Analysis of Netflix Content

```
df3 = df[['release_year', 'description']]
df3 = df3.rename(columns = {'release_year':'Release Year', 'description':'Descrip
for index, row in df3.iterrows():
    d=row['Description']
    testimonial = TextBlob(d)
    p = testimonial.sentiment.polarity
    if p==0:
        sent = 'Neutral'
    elif p>0:
        sent = 'Positive'
    else:
        sent = 'Negative'
    df3.loc[[index, 2], 'Sentiment']=sent

df3 = df3.groupby(['Release Year', 'Sentiment']).size().reset_index(name = 'Total

df3 = df3[df3['Release Year']>2005]
barGraph = px.bar(df3, x="Release Year", y="Total Count", color = "Sentiment", ti
barGraph.show()
```



### Sentiment Analysis of Content on Netflix

